

DATA MINING PROJECT
Master in Data Science and Advanced Analytics

NOVA Information Management School
Universidade Nova de Lisboa

Exploratory Data Analysis for Customer Segmentation at Amazing International Airlines Inc.

Group 44

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Fall Semester 2025-2026

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1. Abstract

This project analyzes Aurora International Airlines' customers behavior and their loyal over time. After cleaning and organizing the data, we found that a small group of frequent, high-value travelers contributes most of the company's revenue. Loyalty status and Customer Lifetime Value (CLV) stood out as the most important factors in understanding customer behavior. We also saw clear seasonal travel trends and mostly solo travelers. These insights will help the airline make better, data-driven decisions.

2. Introduction

Aurora International Airlines works in a very competitive market, where understanding customers and keeping their loyalty are important for long-term success. As the airline industry changes, using data to analyze customer behavior has become an important way to improve relationships and find useful insights about different types of customers. Some travelers look for comfort and high-quality service, while others care more about price or how often they travel. This project focuses on the Business and Data Understanding phases of the CRISP-DM framework. Its main goals are to check if data is consistent, identify any quality issues, and explore demographic, financial, and behavioral information to find meaningful patterns. The analysis helps AIAI understand how customers interact with the company, how they spend and how loyal they are. These findings will support better, data-based decisions and help AIAI create strategies that improve customer satisfaction and overall value.

3. Data Analysis

3.1 Customers

The dataset present in the *DM_AIAI_CustomerDB.csv* file was saved as a pandas dataframe named 'customers'. This dataframe contains information on 16 921 customers of the Amazing International Airlines, with these being described by 20 features that give insight into demographic, behavioral and financial insights into the customer base.

After looking into our dataset, we moved on to a data quality assessment, which revealed a few key issues. Despite, there being no duplicate rows, we found 163 duplicate **Loyalty#** identifiers that affected 363 instances of the dataset. Since they did not belong to the same person (consistent different values for the same features) and constituted a very small portion of our dataset ($\approx 1\%$), we dropped these rows.

All missing values that occurred in the variables **Income** and **Customer Lifetime Value**, occurred simultaneously (and with equal value) in **EnrollmentDate** and **CancellationDate**. Therefore, we assume that it was safe to remove these rows, since these data entries likely represent customers that didn't end up completing any flights. We didn't study this matter any further since these entries correspond to $\approx 0.1\%$ of customers. Missing values in **CancellationDate** were replaced with a placeholder ('not canceled') to show active customers.

We deemed some columns irrelevant or redundant for our analysis like **First Name**, **Last Name**, **Customer Name**, **Country** (with only 'Canada' entries), **City**, **Longitude** and **Latitude**. Although the last two were preserved in a dataframe for future data analysis.

Timeline discrepancies also surged with rows having a **CancellationDate** prior to the value in **EnrollmentDate**. A closer inspection showed that about 75% of these cases consistently showed an offset of around 850 days between the two, suggesting a systematic data entry error. For these cases the dates were swapped to keep the data coherent, while the remaining rows were discarded. After these steps, the dataset was clean and ready to explore.

A curious observation was that about 25% of customers in the data reported an **Income** value of zero. After further study, we found that all these individuals are the exact same ones that fall into 'College' for the **Education** feature. Instead of treating this as an error, we kept these entries under the assumption that they might be current college students that most likely don't have a fixed personal income since their expenses are often supported by their parents, this was supported by their mostly high **Customer Lifetime Value** (CLV).

Finally, a new variable **ProgramDaysDuration**, was introduced to quantify the total number of days that a customer stayed loyal to AIAI's loyalty program, changing the two date fields to one single and continuous variable.

3.2 Flights

The dataset present in the *DM_AIAI_FlightsDB.csv* file was saved as a pandas dataframe named 'Flights'. The initial dataframe had 608 436 rows and 10 features. Most of these variables were numerical, either discrete or continuous (int64 and float64), while **YearMonthDate** was a categorical ordinal variable (object).

To ensure data quality, it was necessary to check for missing values and duplicates. No missing values or duplicated rows were found. However, the **Loyalty#** variable had multiple entries per customer, corresponding to different months. This needed to be addressed before exploring the data, as it could affect summary statistics, averages, and counts. To handle this, new aggregated features were created for each customer: **TotalFlights**, **TotalFlightsWithCompanions**, **TotalDistanceKM**, **TotalPointsAccumulated**, **TotalPointsRedeemed**, **TotalDollarCostPointsRedeemed**, **AvgPointsPerMonth**, **MaxFlightsInMonth**, **MaxPointsInMonth**,

CompanionFlightRatio, **AvgCompanionsPerFlight**, **SoloFlightRatio**, **PointsRedeemedPerFlight** and **AvgDaysBetweenFlights**.

We also kept a copy of the original dataset in the flights original pandas dataframe, which contains necessary information for our future study of flights based on the season that was lost in the last step.

Some anomalies were also detected. The points system had inconsistencies, with 458 customers redeeming more points than they had. To fix this without losing data, all invalid **PointsRatio** were dropped to avoid a biased future analysis. There were also issues with data types in **TotalFlights** and **TotalFlightsWithCompanions**. These were stored as floats, some with decimal places. Since flight counts variables represent discrete values, they were converted to integers using the `int()` function, rounding down fractional values and assigning the correct datatype.

Looking at descriptive statistics, most variables have fairly balanced distributions, with only mild skewness in flights and distance metrics. **TotalPointsRedeemed** shows a strong right-skew, meaning most customers redeem a few points, while a minority redeem very large amounts. **AvgFlightsPerMonth** and **AvgPointsPerMonth** show slight left-skewness, indicating that most customers maintain above-average monthly activity. Finally, **CompanionFlightRatio** is stable and symmetrical around 23%, reflecting consistent behavior in traveling with companions. These distributions and derived features provide a solid foundation for clustering customers based on their travel and loyalty behavior.

4.Results

4.1. Customers

This section presents the results of the exploratory analysis of the customer data, providing a clear overview of the customers demographic and financial features. First, **histograms were used to study the numerical variables** (figure 1). The **Income** histogram is right-skewed, meaning that most customers have a lower income (below 25 000), with only a few earning substantially more. Most of AIAI's customers are low-income individuals, while high-income customers exist but in a smaller fraction. The company might want to focus on affordable, value-driven products or services to cater to the majority, consider segmenting the high-income minority for premium plans or exclusive loyalty programs or use income-based segmentation for targeted marketing campaigns. The **Customer Lifetime Value** also appears heavily right-skewed, with most customers having a relatively low value, while a very small number generate a higher CLV, which suggests that a small portion of customers drive a large portion of revenue. As for the final feature **ProgramDaysDuration**, the distribution appears relatively uniform with customers participating across a wide range of durations, showing a consistent engagement and without a specific cancellation point. **Boxplots** were created to complement the histograms, showing medians, quartiles, and outliers (figure 2). **Income** has a median of about 3 500 with no outliers, confirming most customers have relatively low incomes. The **Customer Lifetime Value** boxplot shows many outliers from roughly 18 000 to over 80 000, highlighting a small group of customers with extremely high CLV values. **ProgramDaysDuration** is fairly symmetrical with no outliers, indicating a uniform distribution.

Bar charts were plotted to examine the categorical variables and compare the frequencies of their categories (figure 3). **Province or State** shows a big part of the customer base is concentrated in Ontario, followed by British Columbia and Quebec, with the remaining provinces contributing with a significantly smaller number of customers. **Gender** is balanced, with similar numbers of male and female individuals. As for **Education**, a very large share of customers has a Bachelor's degree, followed by college. Very few have high school or advanced degrees (Master's/Doctorate). With this in mind, we can infer that the customer base is relatively well-educated. **Location Code** is evenly distributed across Suburban, Rural, and Urban areas. **Marital Status** shows mainly married customers, followed by single and divorced individuals. In **Loyalty Status**, most customers have "Star", followed by "Nova" and "Aurora". **Enrollment Type** is mostly Standard, with few in the 2021 Promotion program. Finally, **IsCanceled** shows that most customers did not cancel (0), with only a small portion having cancellations(1).

Histograms showed that **Education** level has a mixed relationship with Income (figure 4). College-level customers often have no recorded Income, while those with a bachelor's degree show the widest and highest **Income** range. The other categories tend to cluster at lower income levels. **Customer Lifetime Value** and **ProgramDaysDuration** appear relatively uniform across education levels, although bachelor's customers tend to have lower Customer Lifetime Value. When examining **Marital status** (figure 5), single customers have lower **Income** values, while the others are more evenly distributed. **Customer Lifetime Value** is right skewed for all groups, with married customers covering the widest range, and **ProgramDaysDuration** is longest for married customers, followed by single and then divorced. Regarding **LoyaltyStatus** (figure 6), Star customers are the most frequent across all **income** and **CLV** levels, especially at the lower end, which suggests that higher LoyaltyStatus is associated with greater **Income** and lifetime value. **ProgramDaysDuration** is consistent across all loyalty groups. To complement this study, we created **bar plots** (figure 7) with the average Income and CLV by **LoyaltyStatus**. These ended up summarizing the prior conclusions in a more intuitive way. Both metrics show a consistent increase with increasing LoyaltyStatus. Aurora members have the highest average income

(≈40 000) and highest average CLV (≈10 500), followed by Nova, while Star members record the lowest averages.

Subsequently, **Scatterplots** (figure 8) showed that **Income**, **Customer Lifetime Value** and **ProgramDaysDuration** do not have strong correlations with one another, as most customers have low lifetime value regardless of income or program duration. When comparing Income and Customer Lifetime Value (figure 9), it becomes clear that loyalty level plays a more significant role than income in determining customer value. Aurora members, who represent the top loyalty tier, cluster at higher Customer Lifetime Value levels across various income ranges. Nova members appear in mid to high lifetime value ranges but are fewer in number, while Star members dominate the lower range. Overall, income alone does not predict Customer Lifetime Value, and loyalty status emerges as the key factor associated with higher customer value.

We also plotted a **map using the python library geopandas** (figure 10), to evaluate the geographic distributions of customer lifetime values across Canada, with the bubble size representing the number of customers and the colors, indicating the region's average **Customer Lifetime Value**. This revealed a strong concentration in Ontario and British Columbia, with Ontario showing a lower average Customer Lifetime Value despite its bigger customer concentration. Western provinces may not have as much of a big client concentration but show a higher average lifetime value, suggesting that customers in these regions contribute slightly more value on an individual customer basis.

4.2. Flights

Through the graphical analysis conducted in this project, various meaningful relationships between variables in the flights dataset were identified. First, we examined the seasonal relationship in flight activity (figure 11). The results show a clear seasonal pattern: flight activity peaks between June and August, which aligns with the summer holiday period. There is also a noticeable increase in December, most likely associated with holiday travel. A smaller rise is evident in March and April, reflecting spring holiday travel trends.

Regarding annual variation, 2019 registered the lowest flight volume compared with the following years, 2020 and 2021. Flight numbers in 2020 increased moderately, showing the same pattern as the seasonal peaks still happened, however the volume was lower than in 2021, probably because this was one of the most affected sectors by the pandemic. In 2021, a considerable rise in flight activity was observed, which may reflect stabilisation after pandemic disruptions as well as business expansion initiatives by AIAI.

The total distance flown (figure 12) and the number of flights taken with companions (figure 13) follow a similar pattern to overall flight frequency, with clear peaks during major holiday periods. This suggests that social and leisure travel are key drivers of increased flight activity, particularly in the summer and winter holidays, when travel with companions rises noticeably. The slight decline around spring break may indicate that these trips tend to be shorter and undertaken individually or in smaller groups, as spring holidays in Canada are commonly associated with brief leisure-oriented getaways rather than extensive family travel.

Furthermore, the proportional increase in distance flown relative to total flight count indicates that busy seasons are characterized, not only by higher number of flights, but also by more extensive travel. This trend suggests greater passenger demand during these periods, potentially resulting in more frequent and fuller flights.

After examining the seasonality, we analyzed the behavior of loyalty points, both accumulated and redeemed, (figure 14) from which we were able to get some insights. In certain cases, customers appeared to redeem more points than they had accumulated, which is inconsistent with the expected behavior.

One interesting insight taken from the loyalty points analysis (figure 15) is the presence of a positive, although weak, correlation between the number of flights and the total points redeemed. This indicates that customers who travel more frequently tend to redeem a higher number of loyalty points, as it was expected. However, there is a concentration of customers with zero redeemed points with a high number of flights, which indicates that there are also some people who are frequent travelers and prefer to accumulate loyalty points without actively redeeming them.

To gain more insight into airline customers based on their travel frequency and regularity we used two behavioral features: Average Flights per Month and Average Days Between Active Months because these measures capture both how often and how consistently customers fly with AIAI. We applied a rule-based classification to assign each customer to one of five traveler types (Frequent Flyers, Regular Travelers, Seasonal Travelers, Occasional Travelers, and Yearly Travelers) using the thresholds in table 1 two in Appendix.

The resulting scatter plot (figure 16) revealed a clear negative correlation between flight frequency and inactivity gaps: as customers' average days between flying months increase, their average monthly flight count declines. This relationship confirms that our segmentation captures meaningful behavioral differences rather than just a random variation. Distinct clusters emerged, Frequent Flyers (high-frequency, consistent travelers), Regular Travelers (steady, moderate flyers), Occasional Travelers (low frequency but somewhat consistent), and Vacation Flyers/Seasonal Travelers (long inactivity gaps with bursts of travel activity). These insights provide a strong behavioral foundation for further analysis.

5. Conclusion

The exploratory study gave comprehensive insights to both the demographic and behavioural aspects of the AIAI's customers. Throughout this analysis several clear patterns emerged, with the distribution of Income and Customer Lifetime Value (CLV) revealing a significant right-skewness, suggesting that a much smaller portion of high-value passengers ends up contributing significantly to the overall business profitability. Furthermore, the relationship established between Loyalty Status and CLV indicates that higher-tier members (Aurora and Nova) consistently bring more lifetime and monetary value to the airline. This highlights not only the importance of these customers but also the possibility of these offering cluster insights in the future.

The analysis of flight activities shows seasonal patterns that peaked in the summer and winter months which corresponds to major holiday periods. These trends suggest that a portion of AIAI's customers are leisure-oriented. However, this portion probably doesn't represent the majority of passengers but a select few, since on average most customers catch a flight every month and a half.

Flights with companions also increase significantly during these busy times. However, it's also worth noting that despite this increase, in the summer, peak flights with companions are only approximately 27 500, while total flights are around 100 000. This shows a very small ratio of flights with company, with the great majority of customers seeming to be lone flyers (76%).

From a feature engineering perspective, the newly created variables like ProgramDaysDuration, AverageDaysBetweenFlights and CompanionFlightRatio, successfully capture various loyalty insights. These features enhance the interpretability of future clustering models by representing frequency, consistency and engagement more effectively than raw transactional data. Together, they will enable more accurate identification of customer segments with distinct travel behaviors and loyalty patterns.

In this delivery, we successfully classified traveler types : Frequent Flyers, Regular Travelers, Seasonal Travelers, Occasional Travelers and Yearly Travelers based on both industry benchmarks and behavioral interpretations of the data, which were used to define the thresholds for each category. By leveraging AvgFlightsPerMonth and AvgDaysBetweenActiveMonths, we captured not only how frequently customers fly but also how consistently they do so over time. This approach produced a scatter plot that revealed distinct clustering patterns, even with manually set parameters, highlighting the strength of these variables and their potential value for future data-driven cluster analyses.

In the next phase of this project, we will merge the datasets, apply clustering algorithms and formalize these customer groupings. In conclusion, this analysis not only provides a robust technical foundation for clustering, but also offers practical and relevant insights. By aligning analytical findings with business objectives, AIAI can develop a segmentation framework that enhances customer retention, supports targeted marketing and strengthens its competitive position in the airline industry.

This section prepares a solid foundation for the Data Preparation by establishing the analytical roadmap for clustering in the Phase 2 of the deliverables. Propose your clustering approach based on analysis findings, justify feature engineering decisions, and present specific clustering hypotheses. Outline preprocessing strategy and explain expected segmentation outcomes.

6. Bibliographical References

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Baçaõ, F. L, (2025). Data Mining S6.

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Caleiro, A. (2025). *lab06_data_preproces* [Jupyter notebook]. GitHub. [Notebook4](#)

7. Appendix

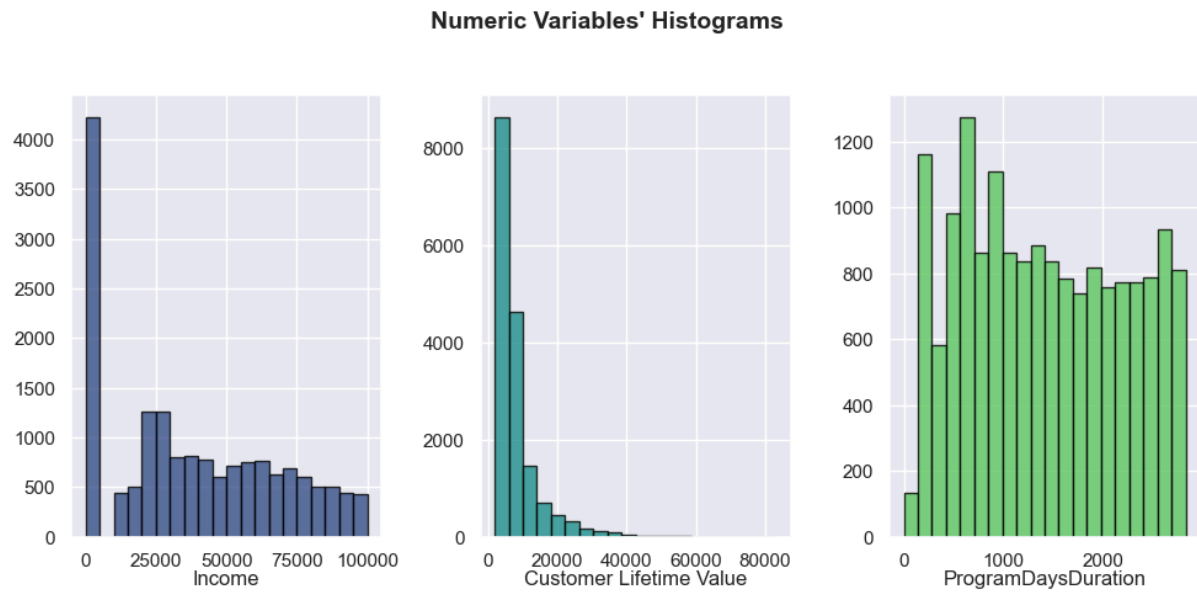


Figure 1 – Histogram of the numeric features of the Customers Data.

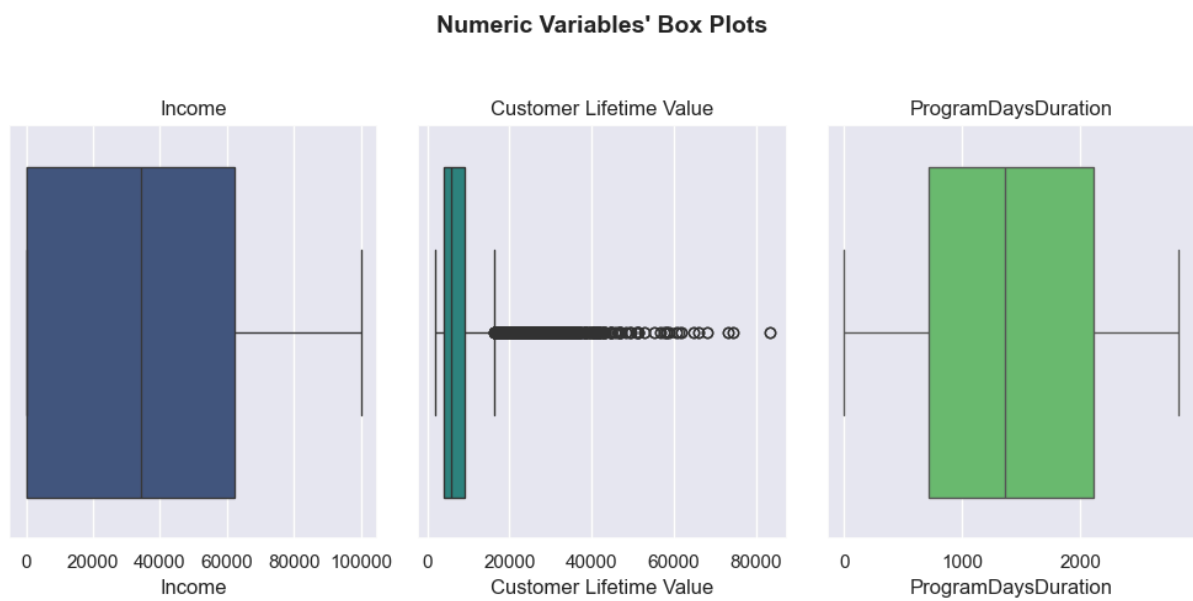


Figure 2 – Boxplot of the numeric features of the Customers Data.

Categorical Variable Distributions

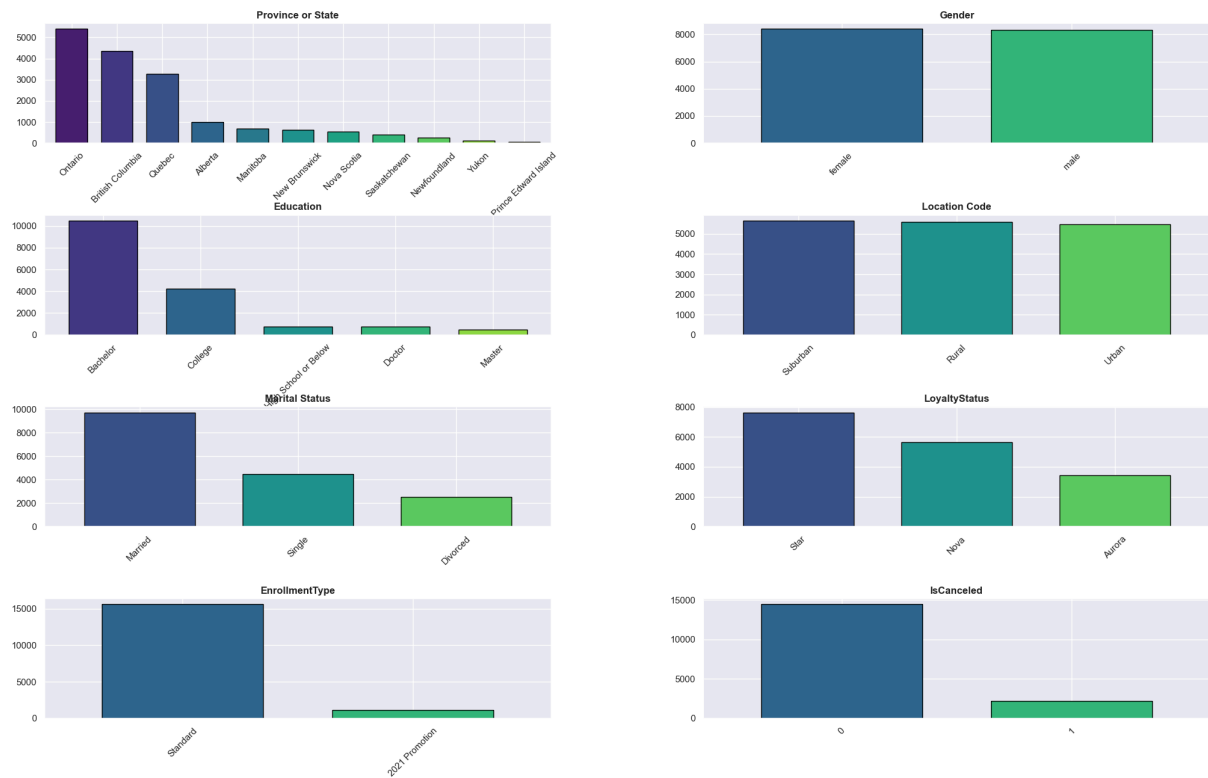


Figure 3– Bar plot of the categorical features of the Customers Data.

Numeric Variables' Histograms

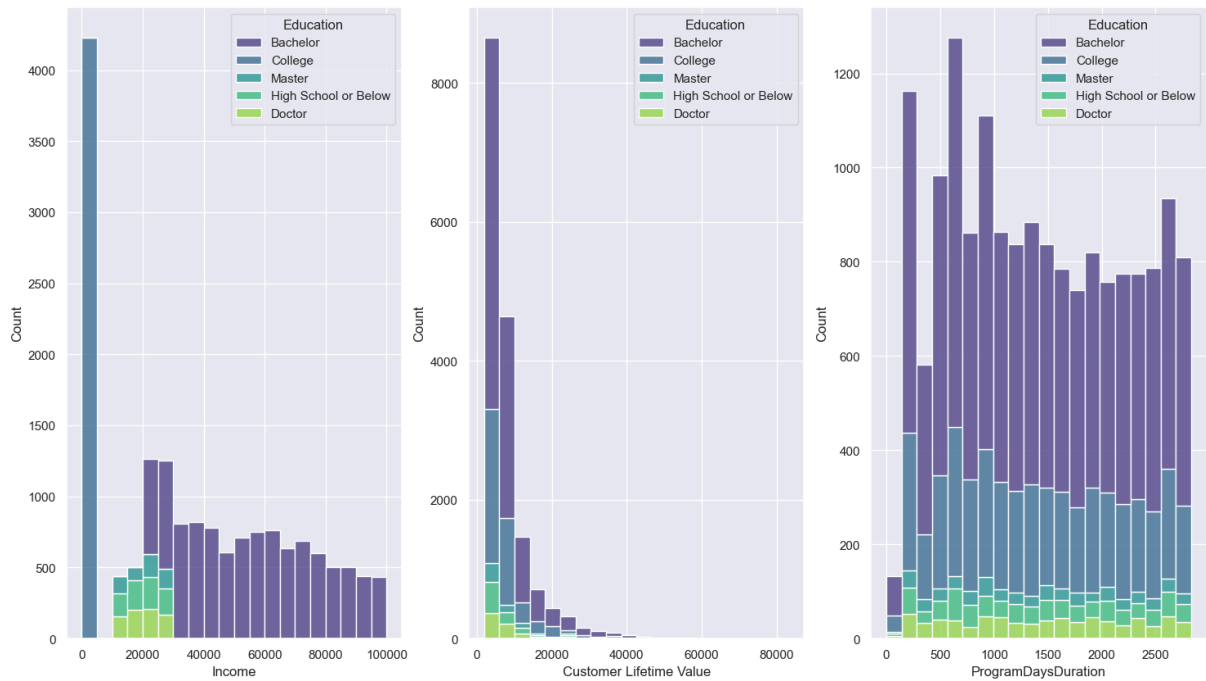


Figure 4 – Histogram of the numeric features with the categorical feature Education.

Numeric Variables' Histograms

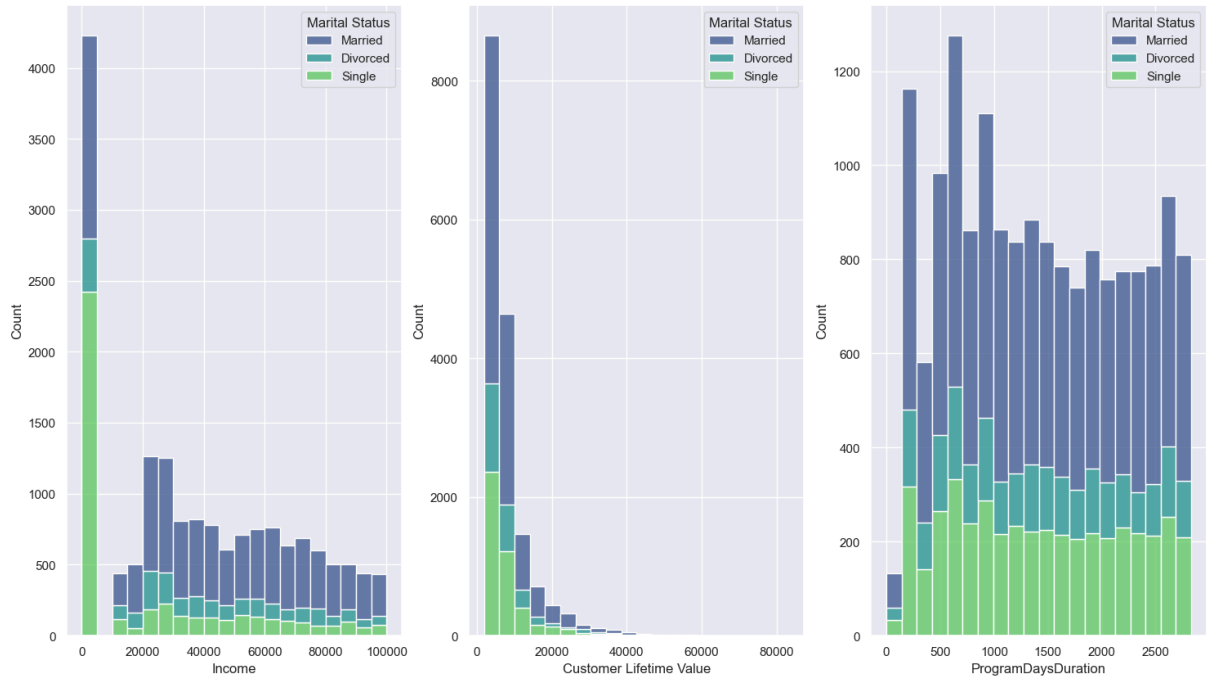


Figure 5 – Histogram of the numeric features with the categorical feature Marital Status.

Numeric Variables' Histograms

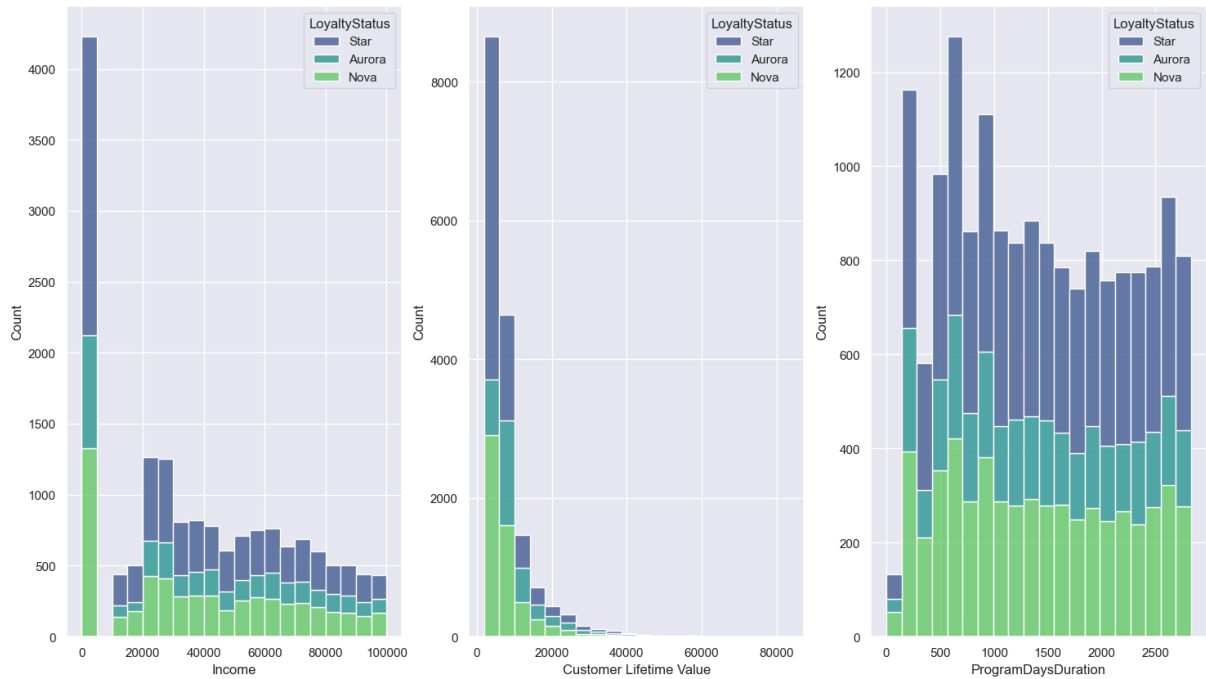


Figure 6 – Histogram of the numeric features with the categorical feature LoyaltyStatus.

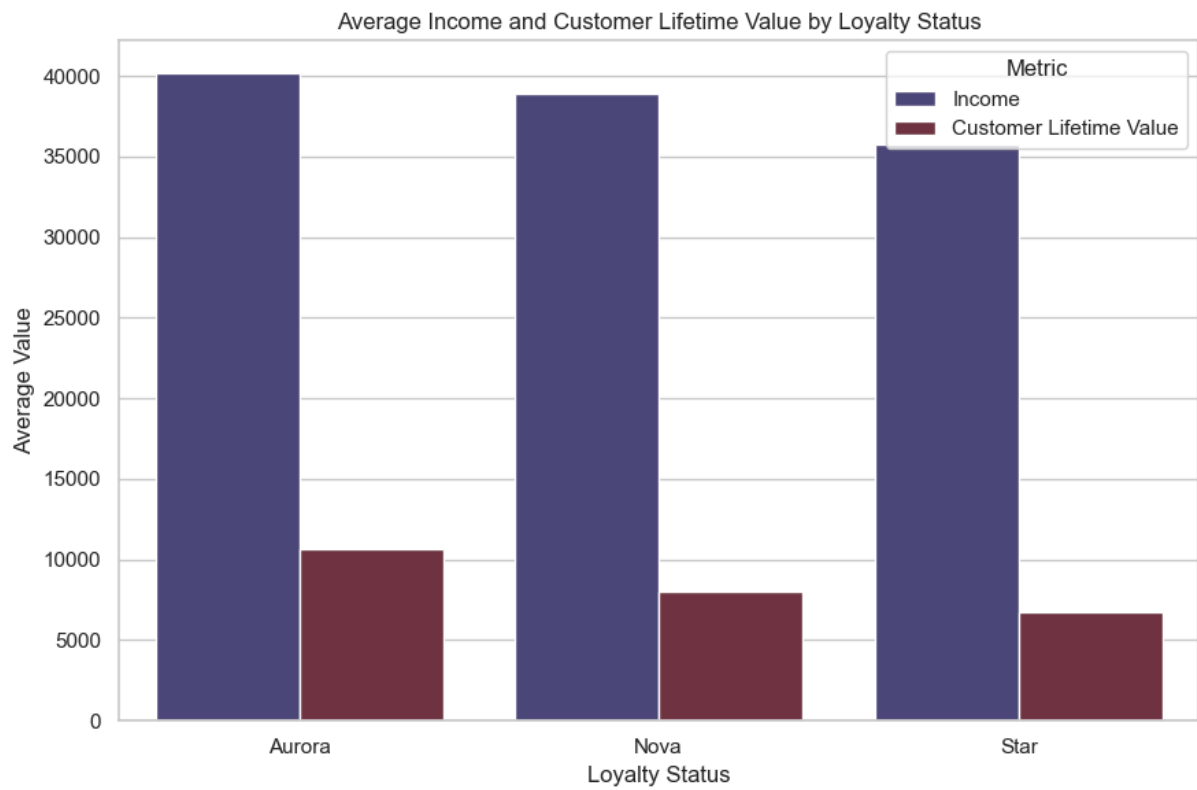


Figure 7 – Bar plot of the average Income and Customer Life Value by Loyalty Satus.

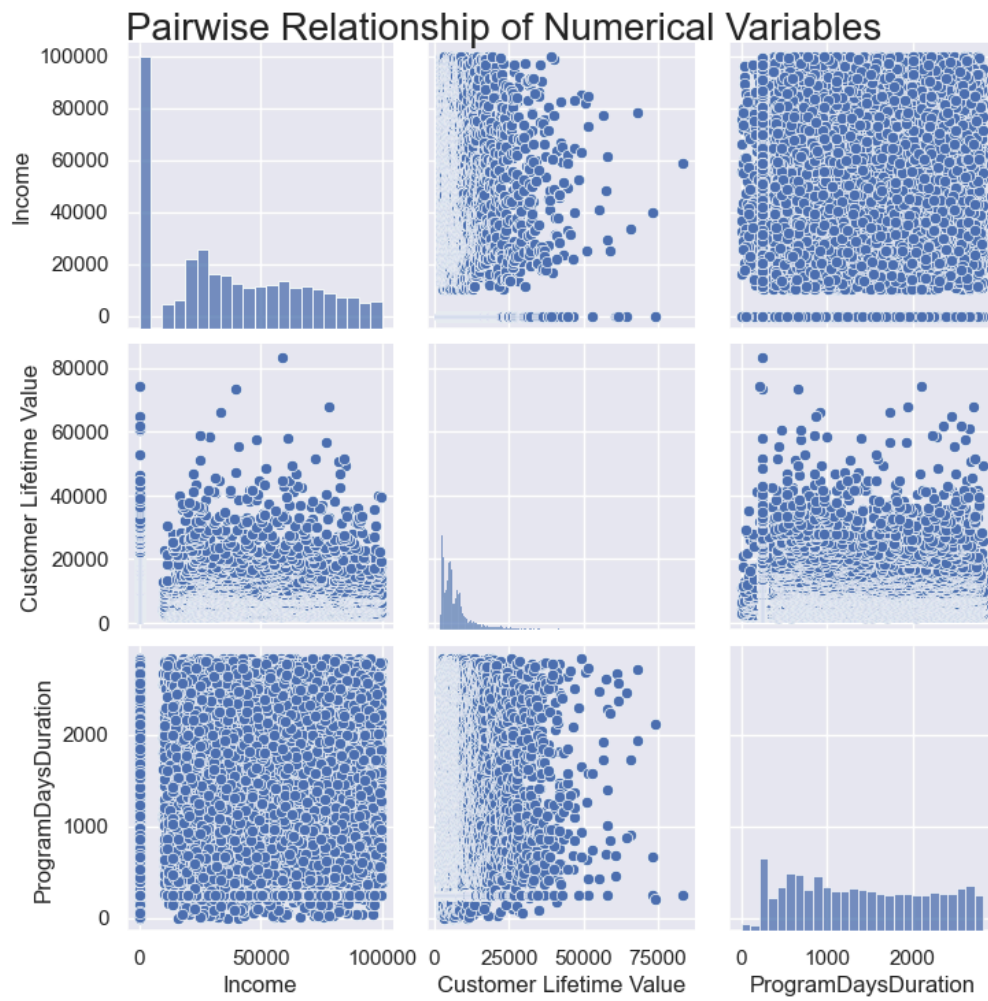


Figure 8 – Scatterplots of the numeric features of the Customers Data.

Boxplot of the numeric features of the Customers Data.

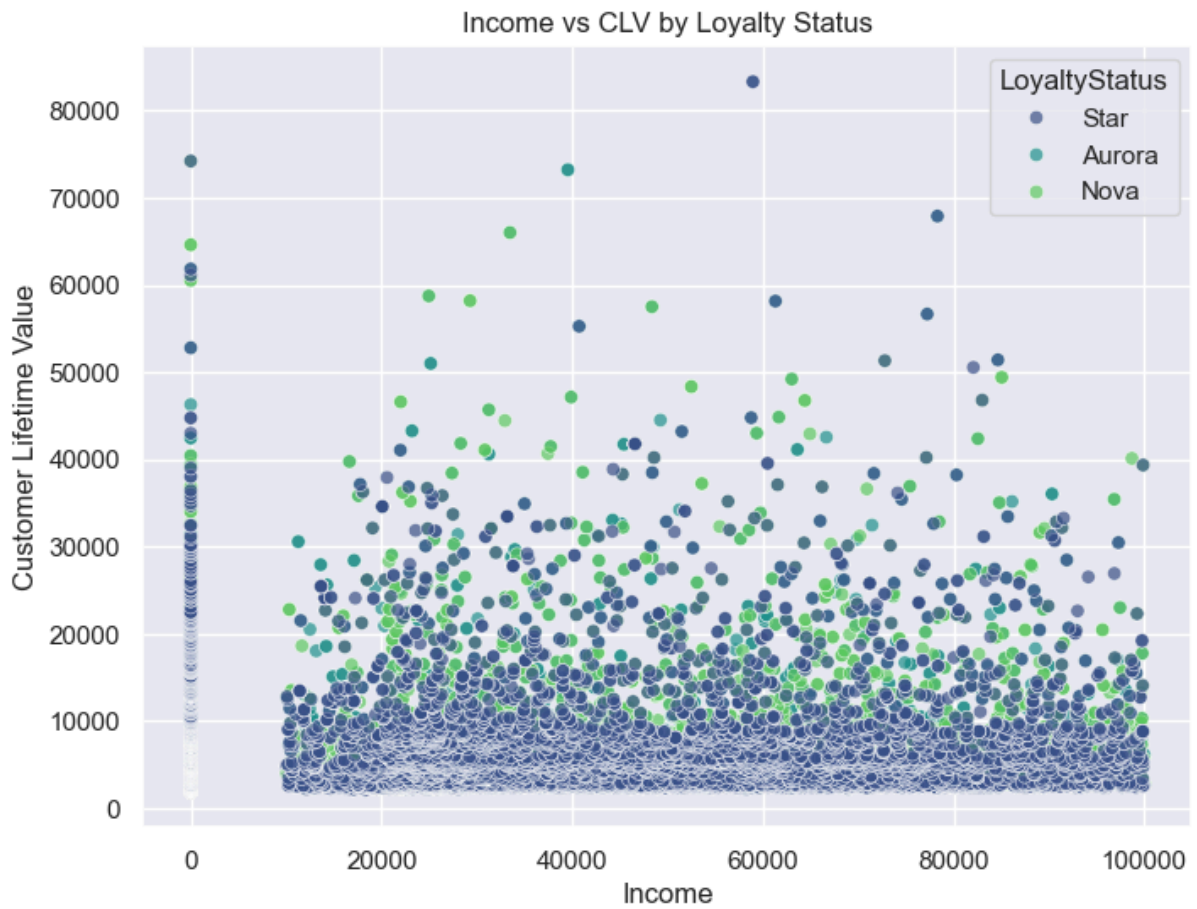


Figure 9 – Scatterplots of the numeric featuresIncome and customer lifetime value by loyalty status.

Geographic Distribution of Customer Value in Canada
Bubble size = # of Customers, Color = Average CLV

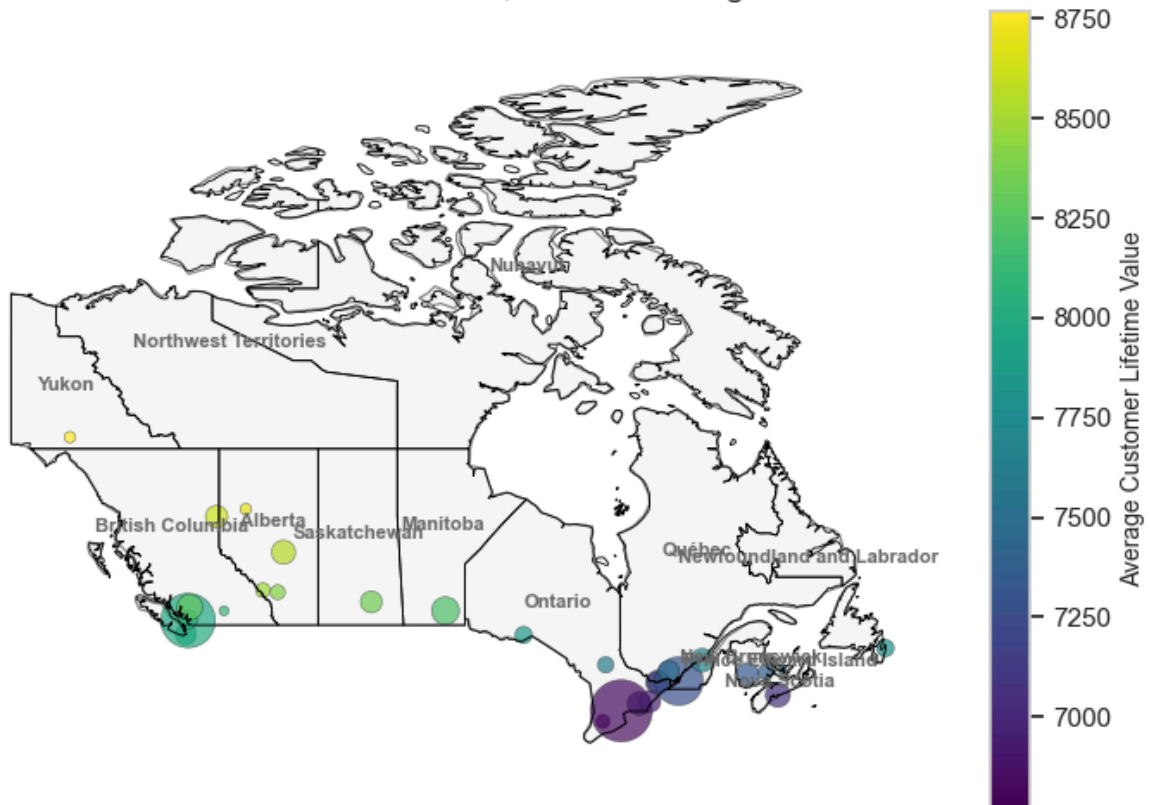


Figure 8 – Geographic distribution of the customer lifetime value.

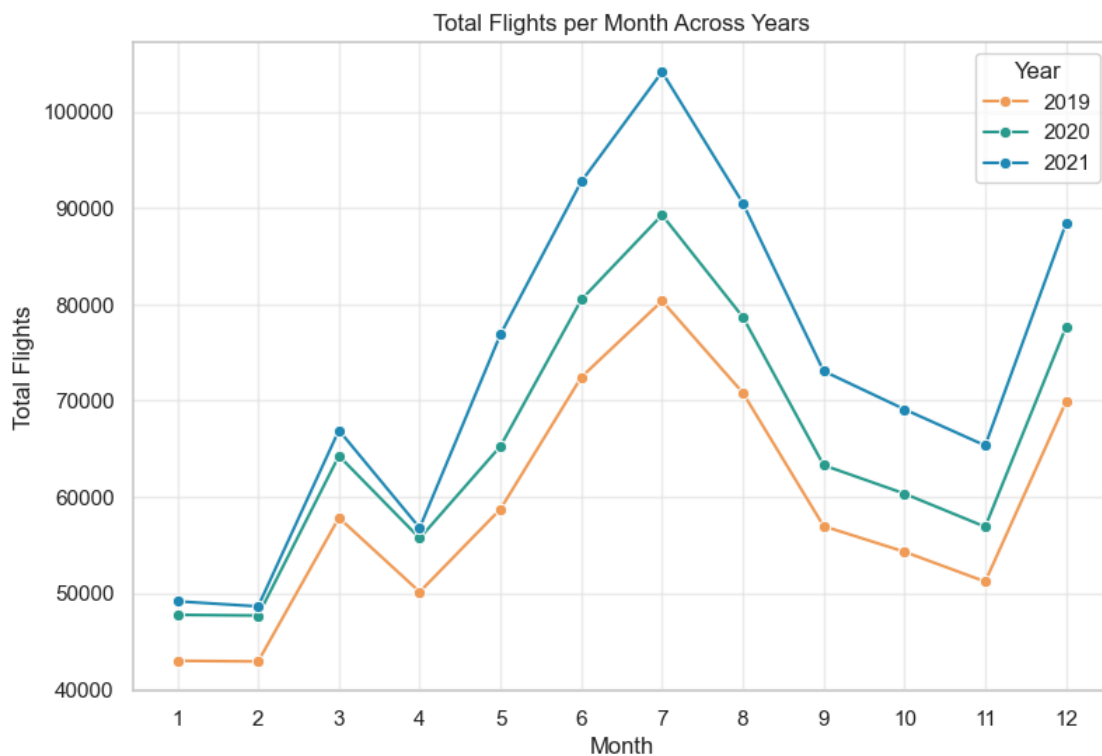


Figure 11 – Total Flights per Month across the Years

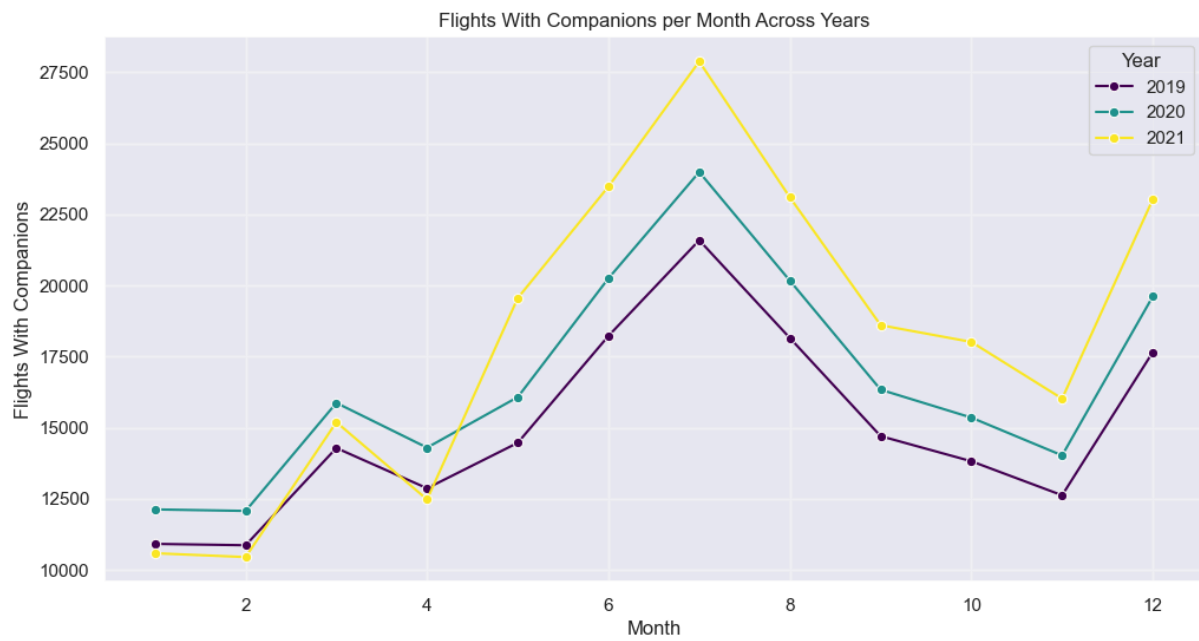


Figure 12 – Flights with Companions per Month across the Years

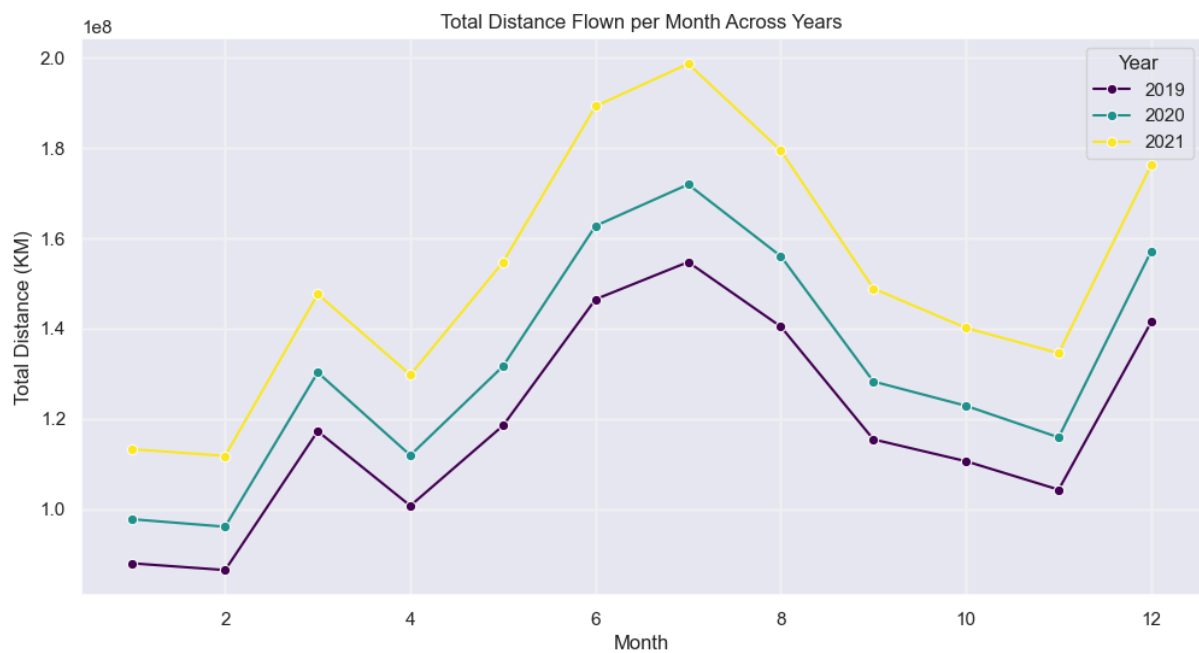


Figure 13 – Flights with Companions per Month across the Years

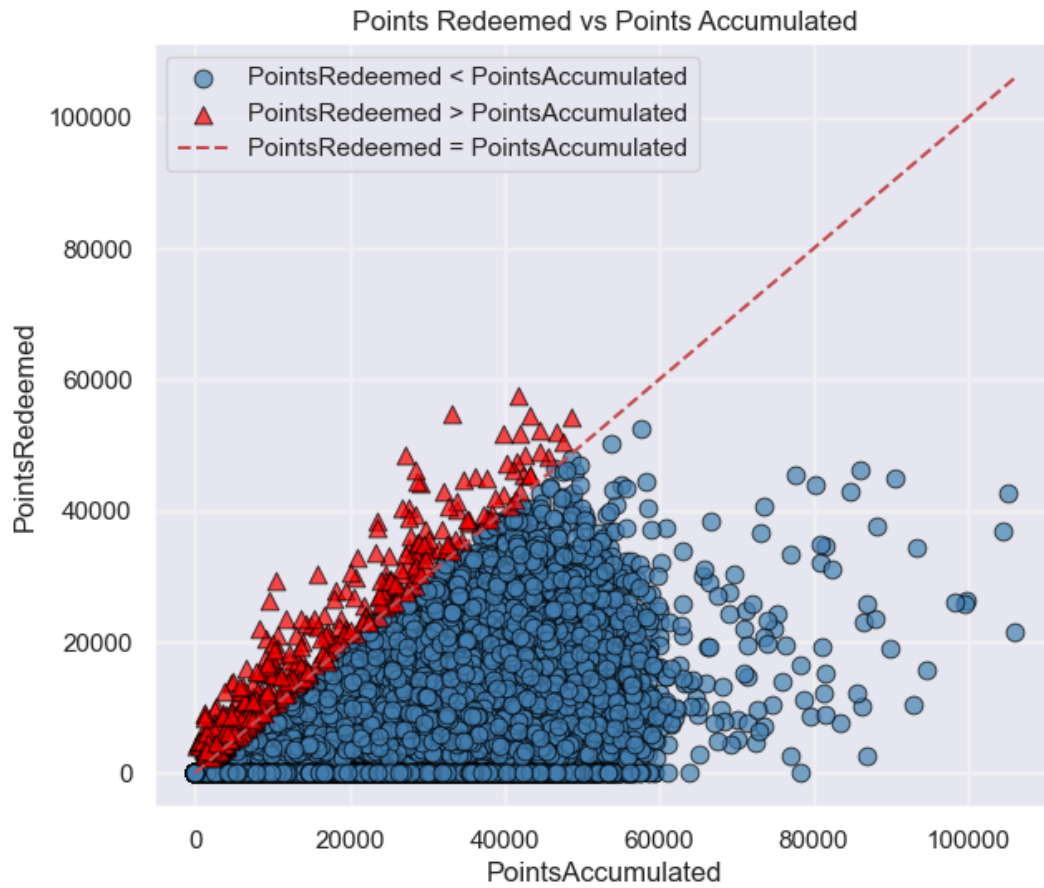


Figure 14 – Points Redeemed vs. Points Accumulated

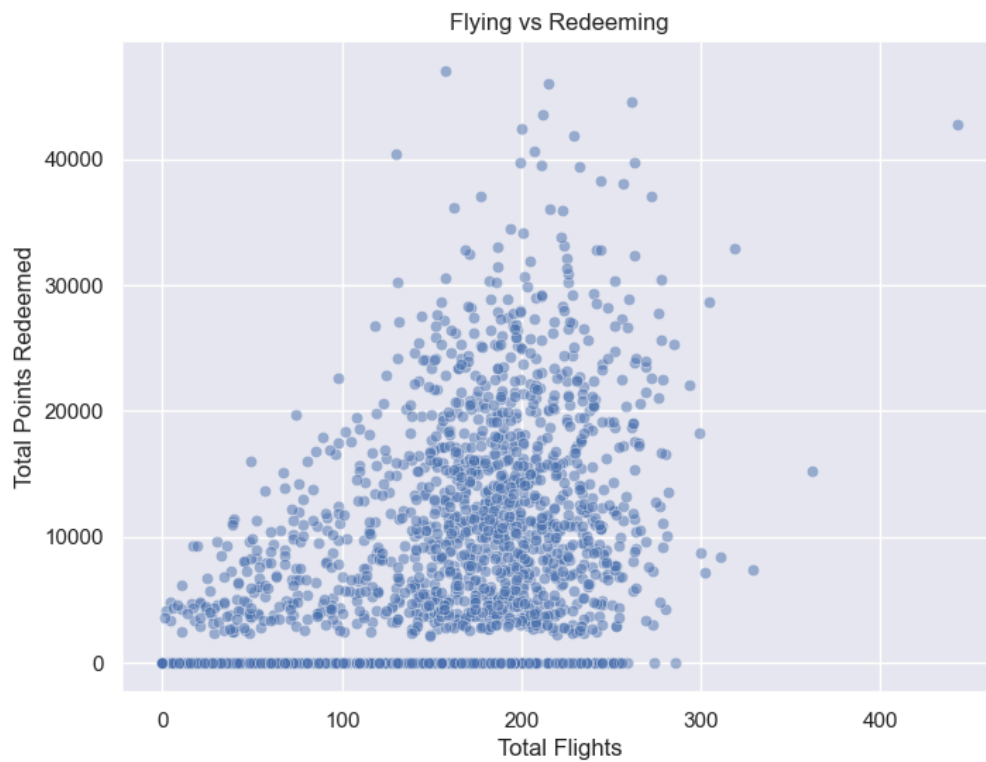


Figure 15 – Flying vs. Redeeming

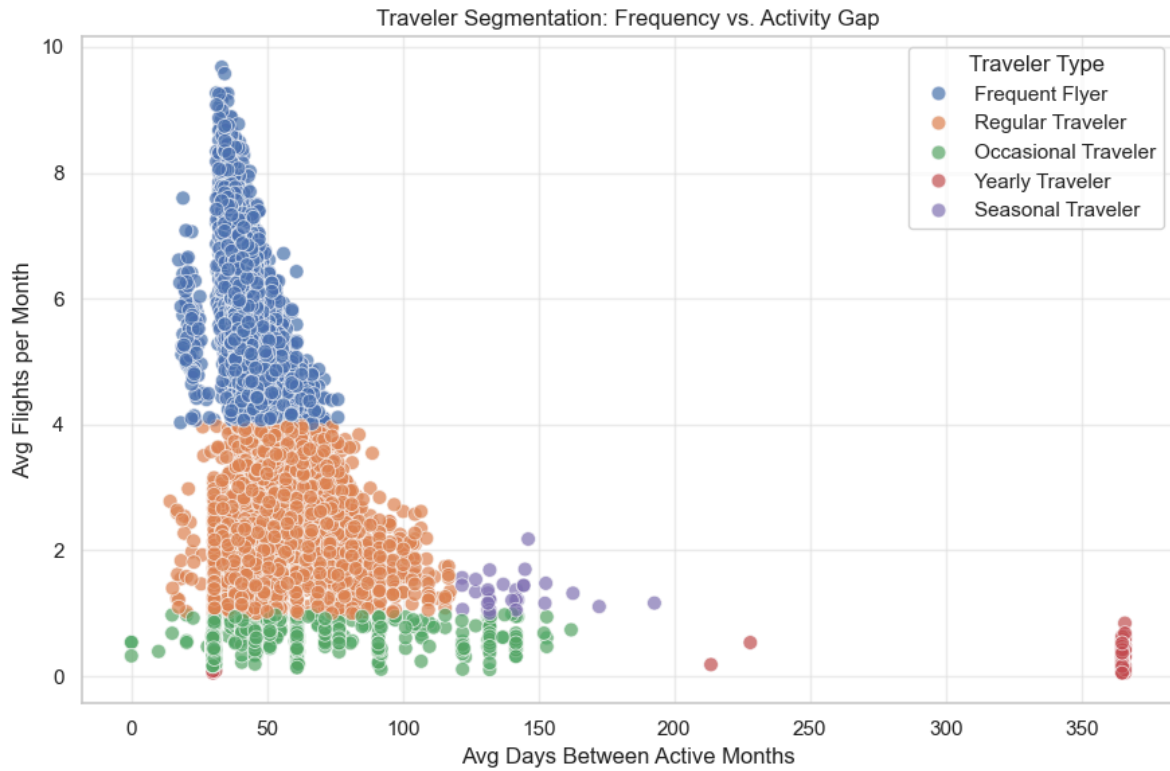


Figure 16 – Traveler Segmentation: Frequency vs. Activity Gap

Table 1 : Thresholds for Traveler Type Classification

Traveler Type	Avg Flights per Month	Avg Days Between Active Months	Interpretation
Frequent Flyer	≥ 4	Any	Roughly one flight per week ($\approx 40\text{--}60$ flights/year); standard industry benchmark for high-frequency travelers and loyalty program elites.
Regular Traveler	1 – 4	< 120	Consistent travel pattern (monthly or biweekly); gaps under 4 months indicate steady, habitual travel.
Seasonal Traveler	1 – 4	≥ 120	Similar frequency when active, but long inactivity gaps (over 4 months); reflects seasonal or holiday travel behavior
Occasional Traveler	0.1 – 1	< 200	Flies a few times per year but not highly seasonal; moderate inactivity between trips.
Yearly Traveler	< 0.1	≥ 200	Very rare flyers, often once or twice a year; highly seasonal or event-driven travel.

7. Annexes

7.1. Annex Statements

AI Usage Statement

- AI tools were used for coding syntax of more complex blocks of code, code debugging assistance and to brainstorm ideas on how to best visualize the data(ChatGPT), and report proofreading and improvement (Grammarly and ChatGPT).

All analytical insights, business interpretations, and strategic recommendations represent original group analysis and thinking.

Contribution Statement

Ana Rita Dias (Student ID: 20250480)

- Pre-processing and cleansing of the flights dataset (Data Wrangling),
- Exploratory data analysis of flights data behaviour (seasonal patterns, loyalty points trends, customer flight frequency analysis),
- Design and development of the poster.

Beatriz Pinto (Student ID: 20250364)

- Pre-processing and cleansing of the customers dataset (Data Wrangling),
- Exploratory data Analysis of customers behaviour,
- Geospatial data integration and analysis,
- Development of interactive dashboards.

Catarina Santos (Student ID: 20250501)

- Pre-processing and cleansing of the customers dataset (Data Wrangling),
- Exploratory data analysis of customers behaviour.

Marta Feital (Student ID: 20250407)

- Pre-processing and cleansing of the flights dataset (Data Wrangling),
- Exploratory Data Analysis of flights data behaviour (seasonal patterns, loyalty points trends, customer flight frequency analysis).

All members contributed to collaborative discussions and ideation as well as for the development and review of the final report.

Responsibility Statement

We, the group members listed above, certify that this report represents our original analytical work and interpretations. While AI tools were used as specified above, all insights, conclusions, and recommendations are the result of our independent analysis and critical thinking. We take full responsibility for the accuracy and quality of this submission.

7.2. Annex: Dataset Information

DM_AIAICustomerDB

Variable	Description
Loyalty#	Unique customer identifier for loyalty program members
First Name	Customer's first name
Last Name	Customer's last name
Customer Name	Customer's full name (concatenated)
Country	Customer's country of residence
Province or State	Customer's province or state
City	Customer's city of residence
Latitude	Geographic latitude coordinate of customer location
Longitude	Geographic longitude coordinate of customer location
Postal code	Customer's postal/ZIP code
Gender	Customer's gender (male, female)
Education	Customer's highest education level (Bachelor, College, etc.)
Location Code	Urban/Suburban/Rural classification of customer residence
Income	Customer's annual income
Marital Status	Customer's marital status (Married, Single, Divorced)
LoyaltyStatus	Current tier status in loyalty program (Star > Nova > Aurora)
EnrollmentDateOpening	Date when customer joined the loyalty program
CancellationDate	Date when customer left the program
Customer Lifetime Value	Total calculated monetary value of customer relationship
EnrollmentType	Method of joining loyalty program

DM_AIAI_FlightsDB

Variable	Description
Loyalty#	Unique customer identifier linking to CustomerDB
Year	Year of flight activity record
Month	Month of flight activity record (1-12)
YearMonthDate	First day of the month for the activity period
NumFlights	Total number of flights taken by customer in the month
NumFlightsWithCompanions	Number of flights where customer traveled with companions
DistanceKM	Total distance traveled in kilometers for the month
PointsAccumulated	Loyalty points earned by customer during the month
PointsRedeemed	Loyalty points spent/redeemed by customer during the month
DollarCostPointsRedeemed	Dollar value of points redeemed during the month